

**PAPER: Sampling Techniques (Theory)**

**Course Code: STAT-305**

**TIME ALLOWED: 30 mins.**

**MAX. MARKS: 10**

**Attempt this Paper on this Question Sheet only.**

### OBJECTIVE

**Q. No. 1: Encircle the correct option.**

1. If \_\_\_\_\_ sampling is done, each item in the population has a chance of being chosen.  
A. Non-probability  
B. Convenience  
C. Probability  
D. None of above
2. What sample statistic is used to estimate a population parameter?  
A. Point estimate  
B. Sampling error  
C. Bias  
D. Non-sampling error
3. The type of sampling technique where each person in the sampling frame has an equal chance of being selected is best described as \_\_\_\_\_.  
A. Systematic Sampling  
B. Stratified Random Sampling  
C. Simple Random Sampling  
D. None of the above
4. To determine the proportion of students at a university who favor the hiring of a new dean, every tenth student is surveyed. What type of sampling is this?  
A. Random  
B. Cluster  
C. Stratified  
D. Systematic
5. \_\_\_\_\_ is a set of elements taken from a larger population according to certain rules.  
A. Sample  
B. Population  
C. Statistic  
D. Element
6. In sampling with replacement, an element can be chosen:  
A. Less than once  
B. More than once  
C. Once  
D. Difficult to tell
7. People who are available, volunteer, or can be easily recruited are used in the sampling method called \_\_\_\_\_.  
A. Simple Random Sampling  
B. Cluster Sampling  
C. Stratified Random Sampling  
D. Convenience Sampling
8. Which of the following will give a more "accurate" representation of the population from which a sample has been taken?  
A. A large sample based on the convenience sampling technique  
B. A small sample based on simple random sampling  
C. A large sample based on simple random sampling  
D. A small cluster sample
9. Probability distribution of a statistic is called:  
A. Sampling unit  
B. Population unit  
C. Sampling distribution  
D. Sampling
10. The difference between a statistic and the parameter is called:  
A. Standard error  
B. Error  
C. Bias  
D. Sampling error



# UNIVERSITY OF THE PUNJAB

Fifth Semester 2018  
Examination: B.S. 4 Years Programme

Roll No. ....

PAPER: Sampling Techniques (Theory)  
Course Code: STAT-305

TIME ALLOWED: 2 hrs. & 30 mins.  
MAX. MARKS: 50

Attempt this Paper on Separate Answer Sheet provided.

## SUBJECTIVE

Attempt all questions.

- Q. No. 2. Write a short note on the following: (5 each)
- Sources of non-sampling errors
  - Inverse Sampling
  - Advantages of systematic sampling
  - Optimum Allocation in Stratified Random Sampling

- Q. No. 3. For simple random sampling, show that the sample mean  $\bar{y}$  is an unbiased estimator of the population mean  $\bar{Y}$  and variance of sample mean is given by (07)

$$V(\bar{y}) = \left( \frac{N-1}{Nn} \right) S^2$$

Use Cornfield approach to prove this.

- Q. No. 4. Compare the values obtained for  $V(p_{st})$  under proportional allocation and optimum allocation for fixed sample size in the following two populations. Each stratum is of equal size. The f.p.c may be ignored. (07)

| Population 1 |       | Population 2 |       |
|--------------|-------|--------------|-------|
| Stratum      | $P_h$ | Stratum      | $P_h$ |
| 1            | 0.1   | 1            | 0.01  |
| 2            | 0.5   | 2            | 0.05  |
| 3            | 0.9   | 3            | 0.10  |

What general result is illustrated by these two populations?

- Q. No. 5. Given the results of a stratified random sample, an unbiased estimator of  $V_{ran}$  which is the variance of the mean of a SRS from the same population is; (09)

$$v_{ran} = \frac{N-n}{n(N-1)} \left[ \frac{1}{N} \sum_{h=1}^L \frac{N_h}{n_h} \sum_{j=1}^{n_h} y_{hj}^2 - \bar{y}_{st}^2 + v_{y_{st}} \right]$$

where  $v_{y_{st}}$  is the usual unbiased estimator of  $V_{y_{st}}$  i.e.  $E(v_{y_{st}}) = V_{y_{st}}$

- Q. No. 6. Show that the variance of the mean of the systematic sample is (07)

$$V(\bar{y}_{sy}) = \left( \frac{N-1}{N} \right) \frac{S^2}{n} [1 + (n-1)\rho_w]$$

where  $\rho_w$  is the correlation coefficient between pairs of units that are in the same systematic sample.